THEORY OF PRODUCTION

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https://sites.google.com/a/nirmauni.ac.in/2hm203-_-eebm_even_2014/

Production: Meaning

Any activity which creates value is production.

In other words, production is transformation of inputs (such as capital, equipment, labour, and land etc) into output such as good or service .

e.g. – transporting sand, operating a jeweller store, drilling for oil, recruiting new employees, designing a system to measure air pollution, producing biscuits, cultivation, trading and so on.

Production Function

Production function express the technological relationship between physical inputs and physical output of a firm under given technology.

A production function may be write as follows

$$Q = f(N, L, K, E, \ldots..)$$

where
Q = output (total product)

N (land), L(labour), K(capital), E(entrepreneurship) , .. are the inputs.

Inputs (Factors)of Production/ Factor Inputs/Factors/Inputs

Factors of production are broadly classified as :

Land:

Anything which is gift of nature and not the result of human effort, e.g. soil, water, forests, minerals. Owner of land is called landlord. Reward of land is called as *rent*.

Labour:

Physical or mental effort of human beings that undertakes the production process. Labour is supplied by the workers. Labour can be skilled as well as unskilled, physical or intellectual. Reward/price of labour is called as *wages/ salary.*

Capital:

Wealth which is used for further production as machine/ equipment/intermediary good. It is outcome of human efforts meaning capital is man-made. Reward of capital is called as *interest*

Enterprise/Entrepreneurship/organisation:

The ability and action to take risk of collecting, coordinating, and utilizing all the factors of production for the purpose of uncertain economic gains. Owner of enterprise is entrepreneur. Reward of entrepreneurship is called as *profit.*

Concept of Time

Alfred Marshal introduced the element of time in production decision. Time can categorize as under:

Market Period or Very Short Period:

Market Period or Very Short Period is a period during which all factors of production and hence cost remains fixed. As such, outputs as well as supply also remain fixed.

Short Run:

Short run a period so brief that the amount of at least one input is fixed. Thus we have both fixed as well as variable factors.

Long Run:

Long Run is a period of time sufficient enough for all inputs (or factors of production), to be variable as far as an individual firm is concerned.

- **The length of time necessary for all inputs to be variable may differ according to the nature of the industry and the structure of a firm.**

Factors (Inputs) of Production: Classification

Factors inputs are classified as fixed and variable.

Fixed factors :

Fixed factors are not related to volume of output. The cost of these factors remain fixed whether output is more or less or even zero.

Variable factors:

Variable factors are directly related to volume of output. E.g. Unskilled labour, raw materials, fuel.

- **Distinction between fixed and variable factors is restricted to short period only. For instance, in the short run plants, machines or equipment are regarded as fixed.**

- **In the long period, all factors are supposed to be variables.**

Concept of Product

There are three concepts of product such as:

- **Total Product**
- **Average Product**
- **Marginal Product**

Total Product :

Total Product refers (TP) to the total volume of goods and services produced by a firm during a given period of time.

$$TP = AP\,xL = \sum \mathrm{MP}$$

Where L is quantity of a factor
AP is the average productivity of a factor.

MP is marginal productivity of a factor .

Average Product:

Average Product (AP) is output (total product) per unit of a factor.

$$\text{i.e. } AP_L = \frac{Q}{L}$$

Where

Q is total product
L is quantity of a factor input.

Marginal Product:

Marginal Product (MP) is rate of change in total product with respect to a factor.

$$\text{i.e. } MP_L = \frac{dQ}{dL}$$

Where

dQ is change in total product
dL is change in quantity of a factor input.

Laws of Production

Theory of production is the study of production functions.

There are two theories of production. Such as:

- **Law of Variable Proportions/ Law of Return to a Factor/ Short Run Production Function**
- **Law of Return to scale / Long Run Production Function**

LAW OF VARIABLE PROPORTIONS/
LAW OF RETURN TO A FACTOR/
SHORT RUN PRODUCTION FUNCTION

Laws of variable proportion studies reaction of output to changes in a variable factor such as labour while others factor inputs are fixed in short turn.

Hence, short-run production function can be written as:

$$Q = f(L, \overline{K}\ \overline{N}, \overline{E}, \ldots)$$

where

Q = output (total product), L is labour and $\overline{K}$, $\overline{N}$, $\overline{E}$ are fixed capital, fixed land, fixed entrepreneurship respectively.

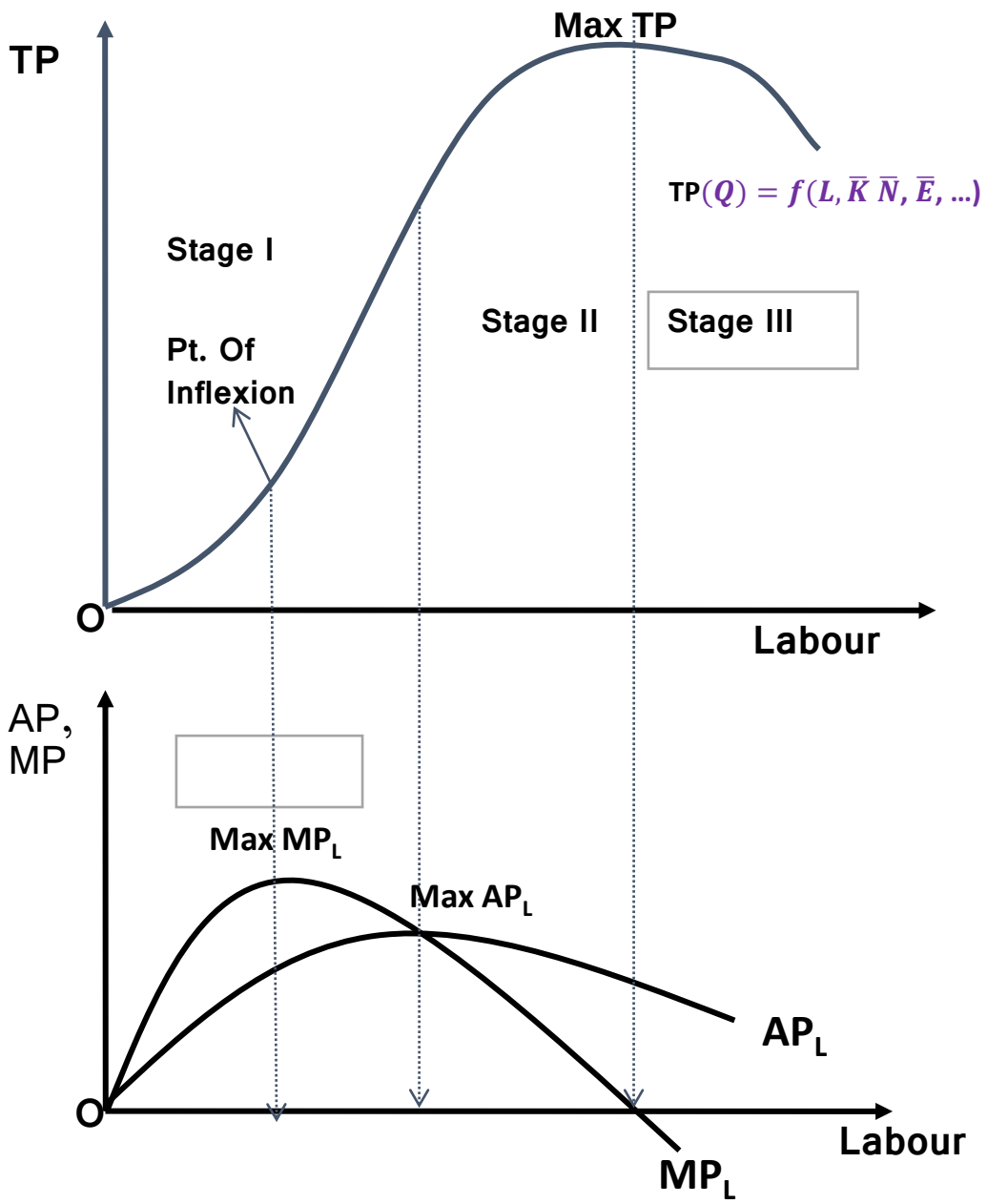
Statement:

Law of variable proportion states as more and more quantities of a factor (say labour) is employed with fixed quantities of other factors in short run, total output increases at

- **an increasing rate (increasing return to a factor),**
- **decreasing rate (decreasing return to a factor),**
- **and finally diminishes after reaching its maximum point (negative returns to a factor) .**

Assumptions:

1. **There is short run**
2. **State of technology is given and unchanged**
3. **Capital is fixed**
4. **Factor-proportion (ratio of variable input to fixed inputs) is variable**



Explanation: Three stages of law of Variable Portions

As the variable input (Labour) is increased while keeping the other factors, behavior of output exhibits three distinct stages. These stages are illustrated graphically as follows.

Stage I: First stage starts from origin and ends at maximum average of the variable factor (AP_L). Total Product (TP) is increasing at an increasing rate up to the point of inflexion at which means marginal product of variable factor (MP_L) attains its maximum point. After point of inflexion, TP increases at a decreasing rate meaning MP_L is falling. MP_L continues to fall until becomes equal to maximum AP_L. At this stage, MP of fixed factor is zero. This stage is called the stage of increasing returns as AP_L is increasing.

Stage II: Second stage starts from maximum AP_L and ends at maximum TP or zero MP_L. TP continues to increase at a decreasing rate meaning MP_L continuing to fall. When TP reaches maxim point, MPL becomes zero. AP_L starts falling from its maximum point. This is stage of diminishing returns because both MP_L and AP_L fall from this stage.

Stage III: Third stage starts from maximum TP or zero MP_L at this stage TP starts falling for which MP_L negative. AP_L continues to fall but would never become equal to zero. This stage is called the stage of negative returns since MP of variable factor is negative.

Stage of Operation: Decision Making by a Rational Producer

A rational producer will choose none of stage I or stage III (which are completely symmetrical). In stage I, the fixed factor is abundant relative to variable factor and hence, MP of fixed factor is negative. In stage III, variable factor is to much compared to fixed factor, and hence MP of variable factor is negative.

Thus, stage I and stage III are called the stages of economic absurdity or economic non-sense, represent non-economic region in production function.

A rational producer will seek to produce at stage II where both MP and AP of variable factor are diminishing returns. Stage II represents the range of rational production decision.

LAW OF RETURNS TO SCALE / LONG RUN PRODUCTION FUNCTION

"Returns to scale" studies the behavior of output or returns when all factor inputs are increased or decreased simultaneously, and in the same proportion in the long-run.

Statement :

When scale (set of all inputs) is expanded in the same proportion, effect on output may take tree forms such as:

- **increasing returns to scale**
- **constant returns to scale**
- **diminishing returns to scale**

The concept of Returns to Scale helps a producer to work out the most desirable combination of factor inputs so as to maximize his output and minimize his production cost.

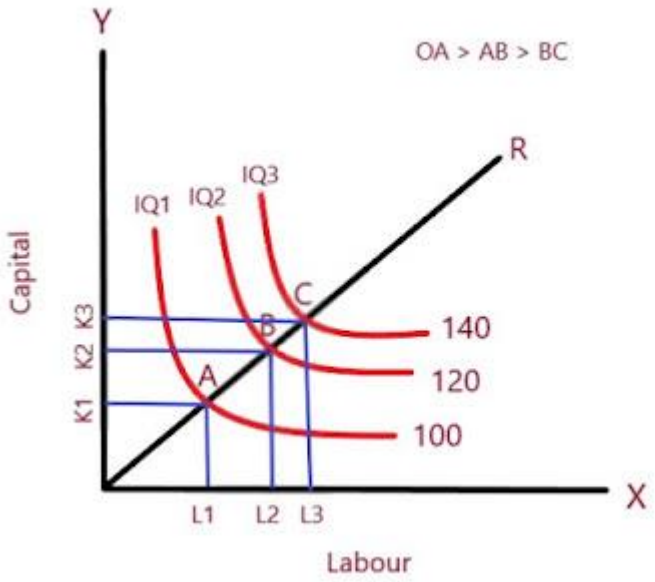
Assumptions:

- All factors (inputs) are variable but enterprise is fixed
- A worker works with given tools and implements
- Technological changes are absent
- There is perfect competition
- The product is measured in quantities
- Factor proportion (ratio of labour to capital) is given

Law of returns to scale using isoquant

1. Increasing returns to scale:-

When a given proportional change in all inputs (i.e. in scale of production) in the long run results in more than proportionate change in output, it is called increasing returns to scale.

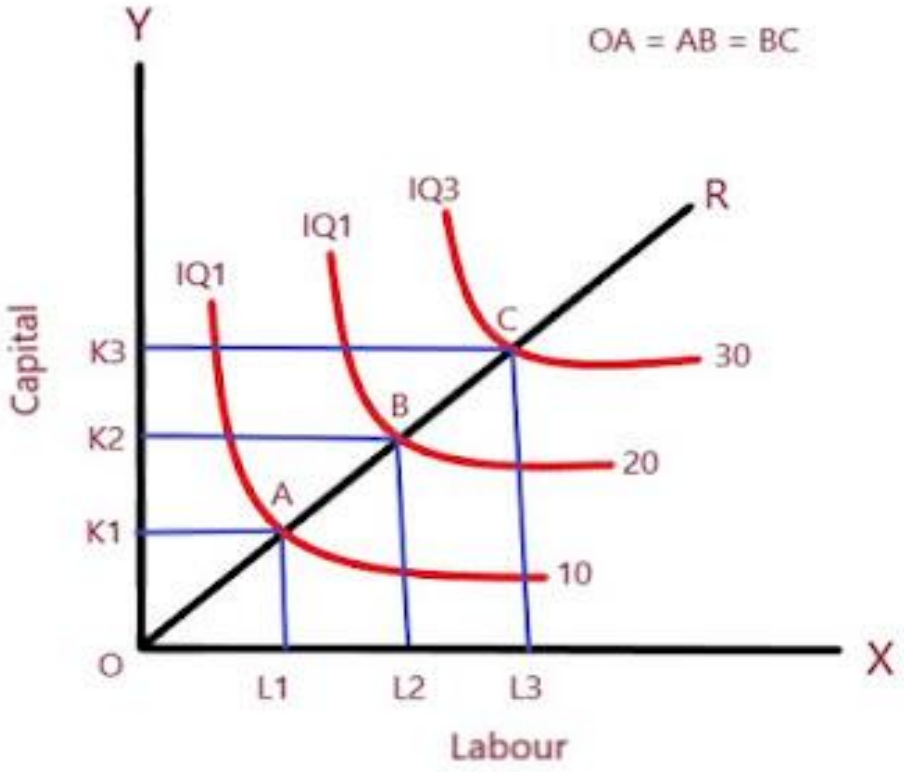


In the above figure, OR is the product line. The various isoquants; IQ1, IQ2, IQ3 are drawn which represent 100, 120, and 140 units of output respectively. A small increase in labor and capital has increased the output by a more amount. I.e. OA>AB>BC is greater than OL1>L1L2> L2L3 and OK1>K1K2> K2K3.

Law of returns to scale using isoquant

2. Constant returns to scale:-

When a given proportional change in all inputs (i.e. in scale of production) in the long run results in the same proportional change in output, it is called constant returns to scale.

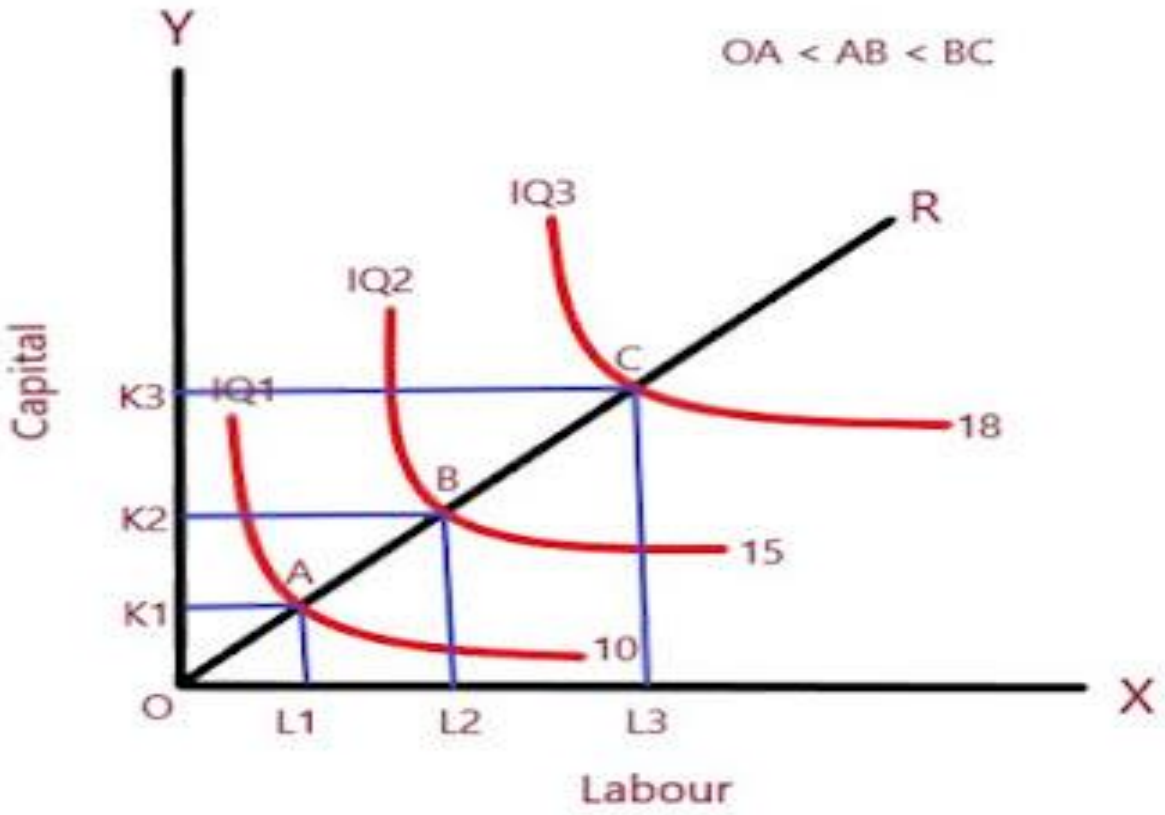


In the above figure, OR is the product line. The various isoquants; IQ1, IQ2, IQ3 are drawn which represent 10, 20, and 30 units of output respectively. Thus, the successive isoquants are equidistant from each other I.e. OA=AB=BC, OL1=L1L2= L2L3 and OK1=K1K2= K2K3.

Law of returns to scale using isoquant

3 Decreasing returns to scale

When a given proportional change in all inputs (i.e. in scale of production) in the long run results in less than proportionate change in output, it is called decreasing returns to scale.



In the above figure OR is the product line. The various isoquants; IQ1, IQ2, IQ3 are drawn which represent 10, 15, and 18 units of output respectively. The successive isoquants are farther from each other. I.e. OA<AB<BC is less than OL1<L1L2< L2L3 and OK1<K1K2<K2K3.

Difference Between Law Variable Proportion & Law of Returns to Scale

Point of Difference	Law Variable of Proportion	Law of Return to Scale
Period of Time	Short Run	Long Run
Variability of Inputs	Only one factor input say labour, is variable others are fixed	All factor Inputs are variables
Factor Proportion	Variable	Fixed
Relationship	Studies how output reacts to changes variable input, while other inputs are kept fixed.	Studies how output reacts to a give proportionate change in all inputs (scale of production)

COST OF PRODUCTION

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COST FUNCTION

Economists have treated cost as a function of output both in short run and long run.

SHORT RUN COST FUNCTION

In the short run, capital, land, factors prices, technology etc remain fixed. Hence, short run cost function can be written as

$$C = f(Q, \bar{K}\ \bar{P}_f, \bar{T}, \ldots)$$

where

C is cost of production

Q is Output (total output)

$\bar{K}$**, are capital and other fixed factor**

$\bar{P}_f$ **is factor prices**

$\bar{T}$ **is given technology**

Thus short run cost function can be written as

$$C = f(Q)$$

LONG RUN COST FUNCTION

The long run cost function can be written as

$C=C(Q, K, T, P_f)$

Where C is cost of production

Q is output
T is technology
K is capital
P_f is factor prices

In studying the relationship between long run cost and level of output in a two dimensional diagram, we keep technology and output as constant.

Thus , long run cost function, traditionally, is written as

$$C= C(Q)$$

CONCEPT OF COSTS(SHORT RUN)

Total Fixed Cost (TFC):

Total Fixed Cost (TFC)/ Fixed Costs are the expenses on fixed factors of production. Fixed cost do not vary with output. They remain the same what ever be the output in short run.

They typically include rents, salaries of permanent employees, insurance, depreciation etc.

Total variable cost (TVC)

Total variable cost (TVC) /Variable costs are expenses on variable factors of production. Variable costs do vary with output. As output changes, TVC also changes.

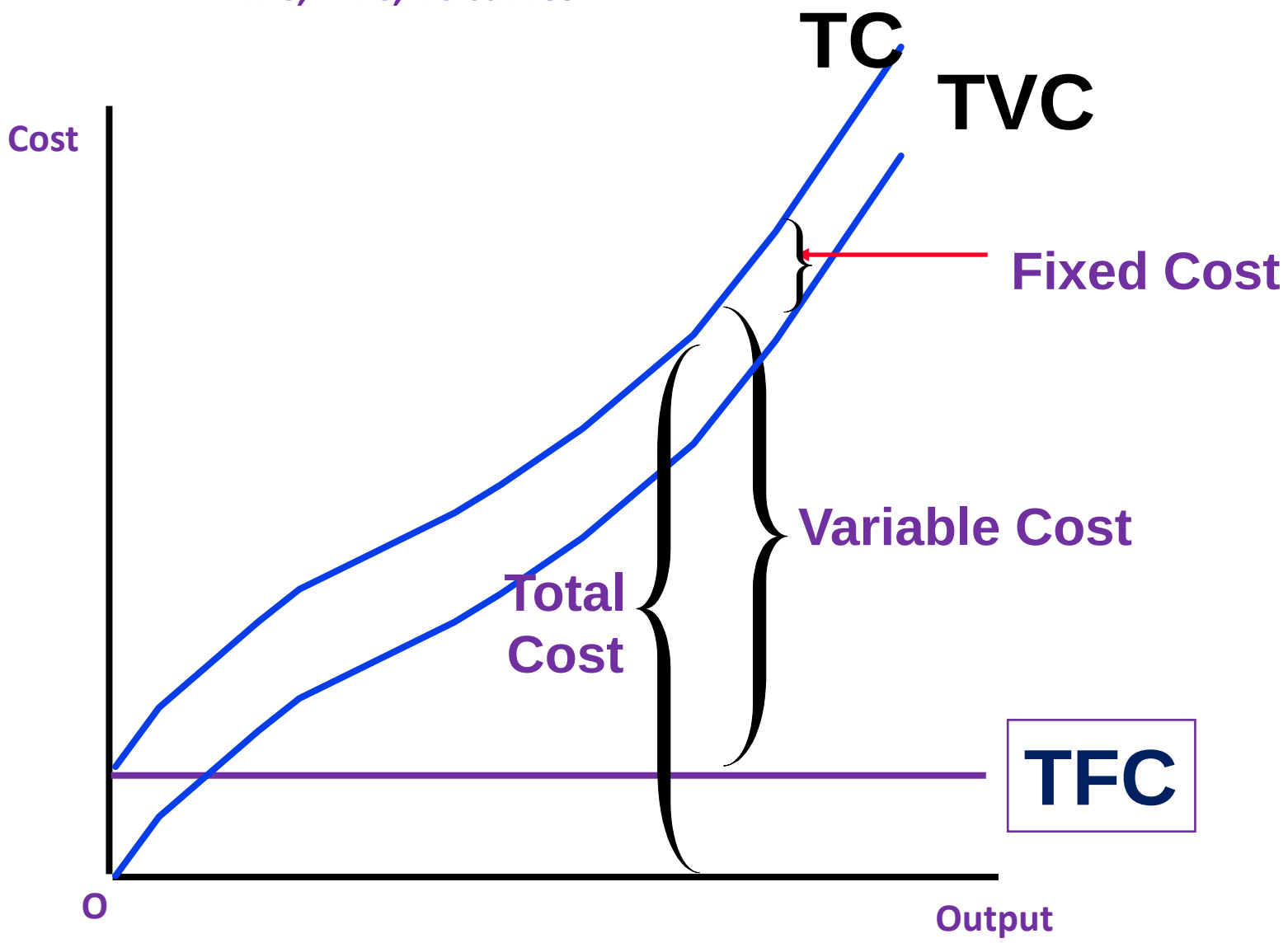
Examples of typical variable costs include fuel, raw materials, and casual labour costs.

Total Cost:

Total cost is the sum of fixed and variable costs at each level o output.

i.e. TC=TFC+ TVC

TFC, TVC, TC curves



Average Fixed Cost (AFC) : ***Average fixed cost is the fixed cost per unit of output.***

$$\text{i.e. } AFC = \frac{TFC}{Q}$$

Where Q is output.

Average Variable Cost (AVC) : Average variable cost is the variable cost per unit of output.

$$\text{i.e. } AVC = \frac{TVC}{Q}$$

Where Q is output.

Average Cost (AC) : *Average cost is the cost per unit of output.*

$$\text{i.e. } AC = \frac{TC}{Q}$$

$$= \frac{TFC + TVC}{Q}$$

$$\boxed{\text{Or, } AC = AFC + AVC}$$

Where Q is output.

Marginal cost (MC) : *Marginal cost is the extra cost for producing an additional unit of output.*

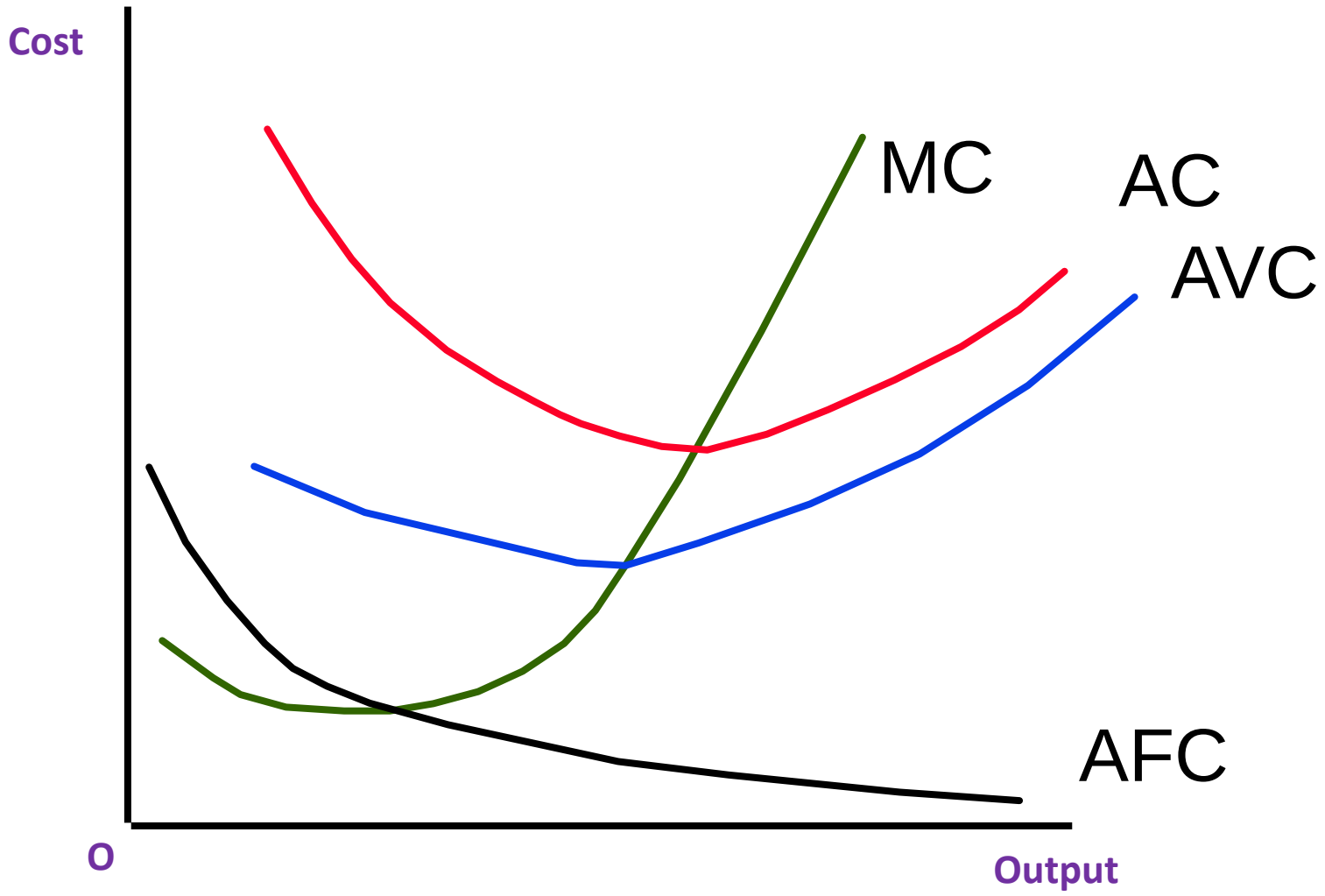
$$\text{i.e. } MC = \frac{d(TC)}{dQ}$$

$$MC = \frac{d(TFC + TVC)}{dQ}$$

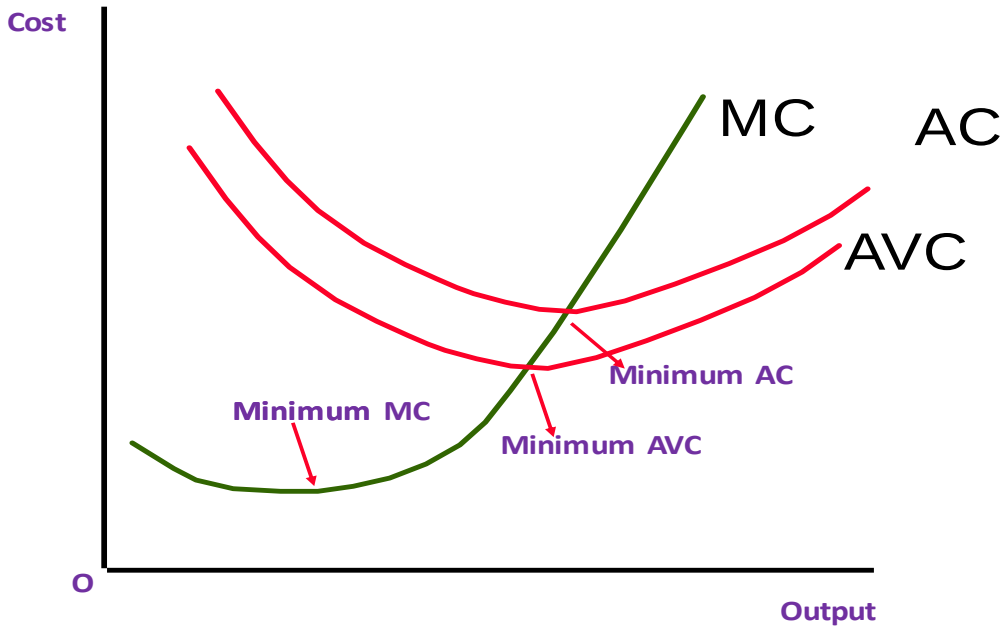
$$MC = \frac{d(TVC)}{dQ}$$

It is important to note that marginal cost is derived solely from variable costs, and not from fixed costs.

AFC, AC, AVC, MC curves



RELATIONSHIP BETWEEN AC AND MC



- ❑Minimum MC comes before, Minimum AC.
- ❑When AC falls, MC < AC
- ❑When AC increases, MC > AC
- ❑When AC is at its minimum, MC equals AC. i.e. MC=Minimum AC.

AVERAGE FIXED COST (AFC) FALLS WITH INCREASE IN OUTPUT

AFC curve is downward sloping from left to right.

REASONS

As total fixed cost is divided by an increasing output, AFC will continue to fall. AFC may be very close to zero, but will never be equal to zero.

AVERAGE VARIABLE COST CURVE , AVERAGE COST CURVE , AVERAGE COST CURVE are IS U-SHAPED

With the increase in output each of these curves will slope down at frist from left to right, then reach a minimum point, and rise again.

'U' shaped AVC , MC, AC curve can be attributed to the principle of variable proportion.

AVC = TVC/Q = (L W) /Q = w/(Q/L) = w/(AP)

MC = d(TVC)/dQ= d(Lw)/dQ= w(dL/dQ) = w /(dQ/dL) = w /MP

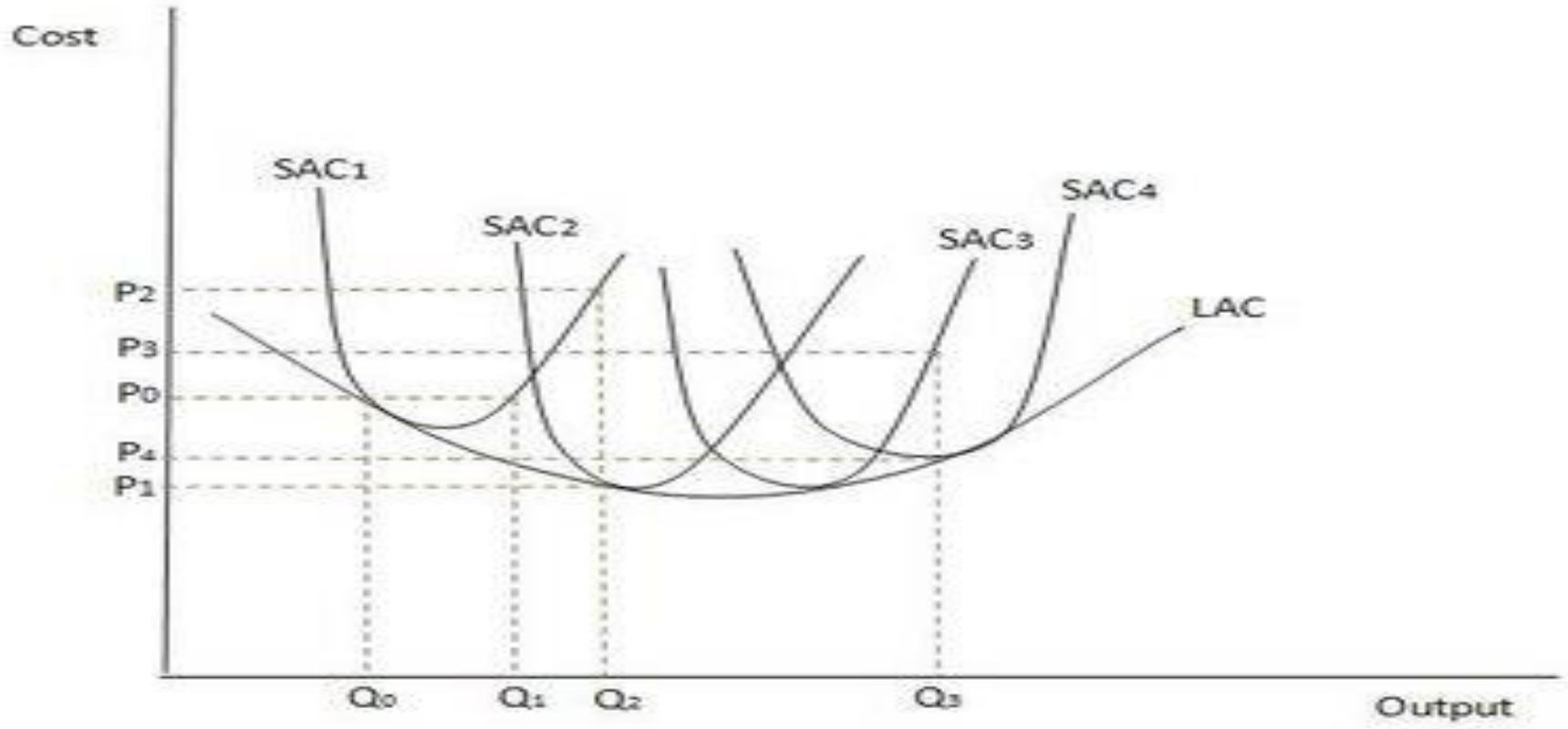
LONG RUN AVERAGE COST FUNCTION
Or
LONG RUN AVERAGE COST (LAC) CURVE

The long run cost function/LAC curve depicts the least possible average cost for producing various levels of output when inputs of production are variable (in the long run).

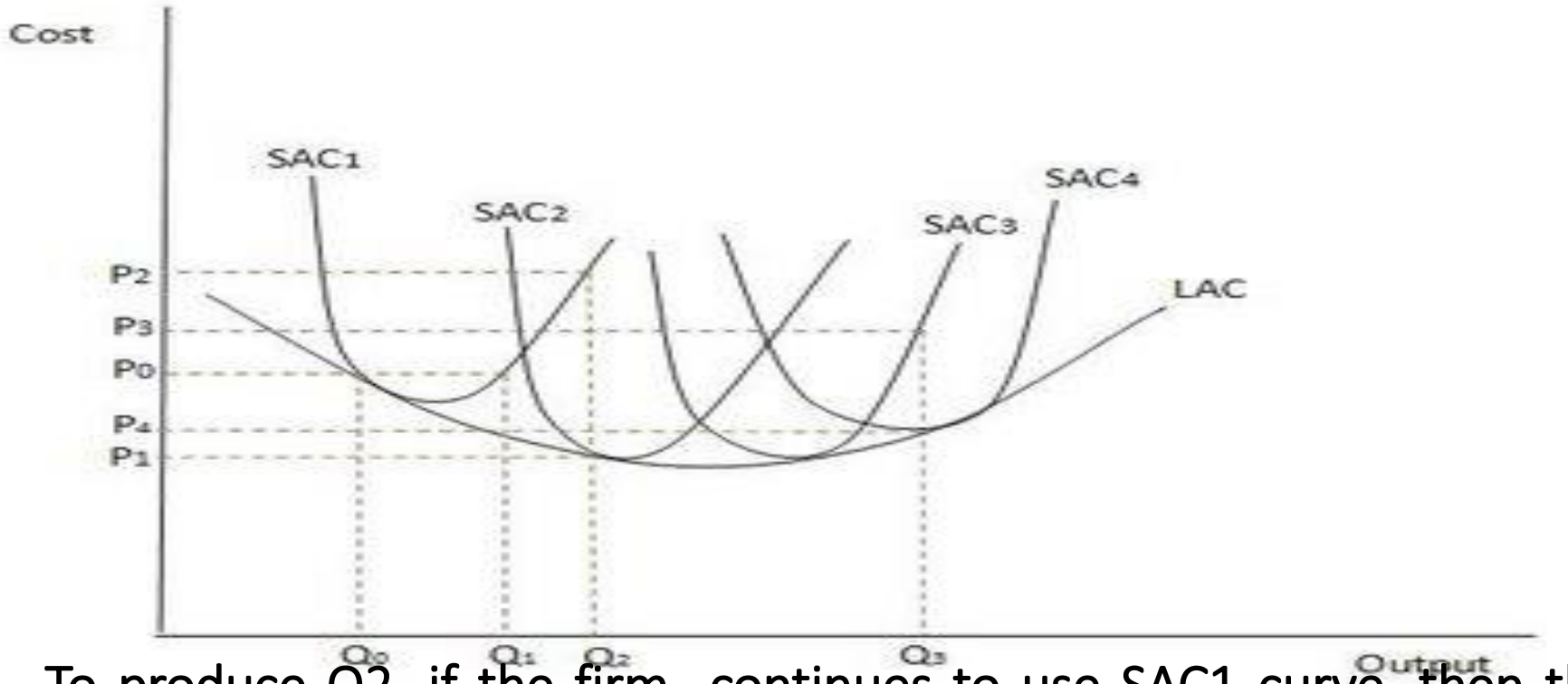
The LAC is constructed from a series of short run average cost curves associated with a series of different output levels. In short run, capital (say plant) is fixed, and labour is variable. As such, each of the short run average cost (SAC) curves corresponds to a particular plant. When a new plant is installed, the SAC curve shifts to a new location.

The LAC curve is the envelope of an infinite number of SAC curves, with each SAC curve tangent to the LAC curve at a single point corresponding to a particular output level.

Let us consider the short run curves given by SAC1, SAC2, SAC3 and SAC4 as in the following figure.



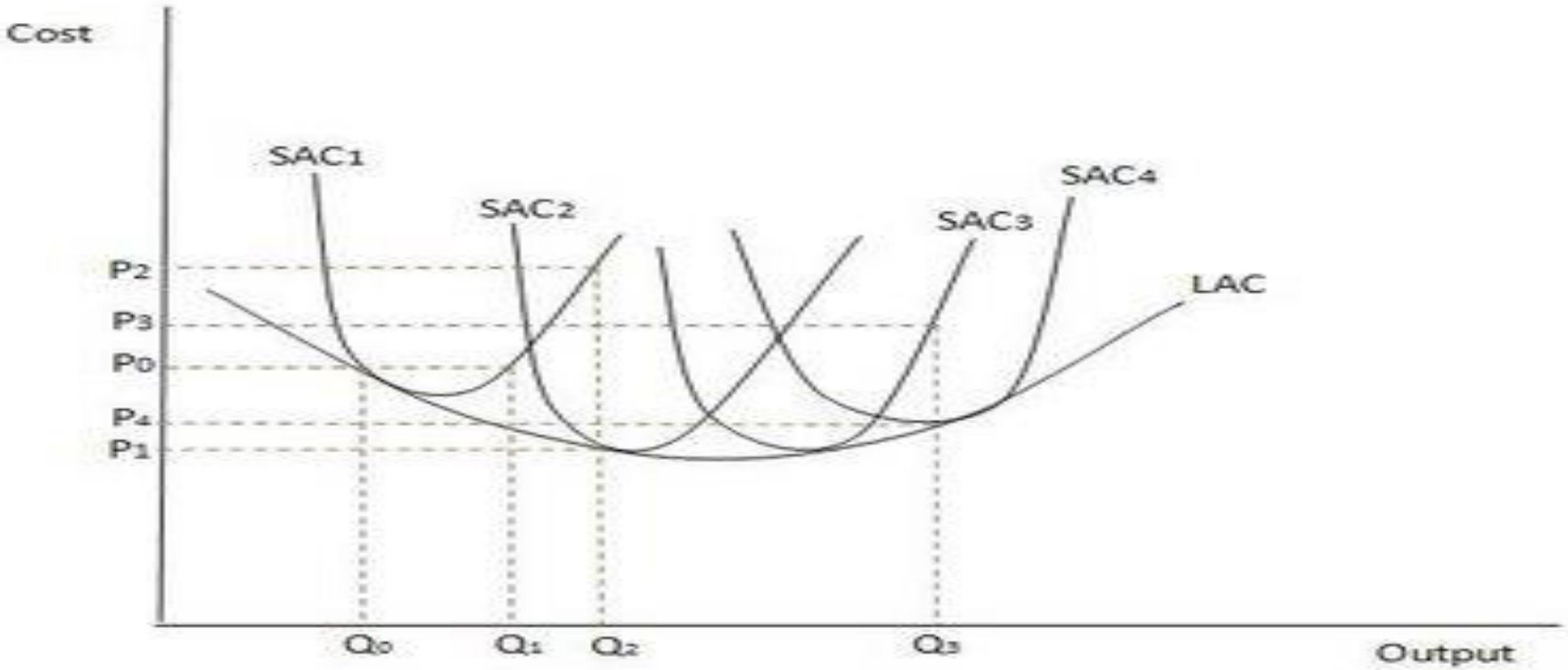
To produce Q0, the firm will use SAC1 curve where cost is P0. To produce a higher level of output at Q1, the firm will again use SAC1 curve at which the cost remain the same at P0.



To produce Q2, if the firm continues to use SAC1 curve, then the cost will increase to P2(P2>PO). So, it would be better for the firm to bring a second plant into the production with cost curve SAC2. Here, cost of producing Q2 would be P1, much less than P2.

Similarly, to Produce Q3, the firm can either use SAC3 or SAC4 curve. If it uses SAC3 curve, the cost of production would be P3 which is higher than P4 at SAC4. Hence, the firm will use SAC4 curve, as it has low cost(P4).

LONG RUN AVERAGE COST FUNCTION (contd)



The above description confirms that, in long run, a firm will use the level of inputs that can produce a given level of output at the lowest possible average cost. The envelope relationship explains that at the planned output level, SAC equals LAC; but at all other levels of output, SAC is higher than LAC. This is reflected in LAC curve which is always be less than or equal to short-run average cost.